Mechatronic Sensors Trainer

Fundamental Labs – Ultrasonic

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For more information on the solutions Quanser offers,
please visit the web site at: http://www.quanser.com

Quanser Consulting Inc.
119 Spy Court
Markham, Ontario
L3R 5H6, Canada
info@quanser.com
Phone : 19059403575
Fax : 19059403576
printed in Markham, Ontario.

Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Ultrasonic Lab

Why Explore Ultrasonic Sensors?

Distance measurement is a core requirement when performing tasks such as obstacle avoidance, presence detection, and surveying. Ultrasonic sensors, which detect distance, are found in applications such as industrial automation in production lines, automotive systems for blind spot detection, and processing plants for liquid level detection. Understanding the working principles of ultrasonic sensors helps us leverage their advantages and work within their limitations.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**026_ultrasonic**

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of the **Ultrasonic Sensor**. The lab will guide you through understanding measurements from an ultrasonic sensor.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.